

Peh Hoe Khim Marcus

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WORK EXPERIENCES

TikTok, Software Engineer

Apr 2024 - Present

- Improved Online Query Engine scalability to sustain >2M peak QPS at ~30ms P99 with >99.99% SLA by optimizing cache access patterns
- Reduced data readiness time by 73%, from 5.5 hours to 1.5 hours, by re-architecting Hive-to-ClickHouse data loading
- Designed and built a multi-region, on-demand data pipeline from scratch for cohort-level Creator Analysis, reaching 30%+ user penetration
- Expanded Data Suite query capabilities by building a visualization engine supporting direct ClickHouse queries and configurable virtual columns

ByteDance, Backend Software Engineering Intern

Aug 2023 – Dec 2023

- Integrated internal approval tooling into the existing approval workflow, improving processing efficiency by 30%
- Improved channel rating API performance by 90%+ by precomputing data and parallelizing category calculations with Go goroutines

Google, Software Engineering Intern

May 2023 - Aug 2023

- Refined onboarding verification flow on Google Pay's merchant application (Android)
- Reduced QR code loading time by 93.5%, from 1.38s to 0.09s, by eliminating unnecessary RPC calls
- Lowered proximity verification range by 99.5% from 2km to 10m by leveraging Bluetooth Low Energy

GovTech, Software Engineering Intern

May 2022 - Jul 2022, Dec 2022 - Jan 2023

- Reduced web page loading time by 10% by refactoring frontend call logic and removing unnecessary synchronous API calls
- Strengthened security by implementing Two-Factor Authentication, mitigating unauthorized access risks
- Improved server query response time under load by 200x by restructuring API request handling and reducing redundant database queries

EDUCATION

Master of Science in Computer Science

Aug 2024 – Present

Georgia Institute of Technology

GPA: 4.00 / 4.00

Bachelor of Computing in Computer Science (Honors)

Aug 2020 – Jan 2024

National University of Singapore

GPA: 4.78 / 5.00 (Highest Distinction)

PROJECTS

Trading Algorithm Framework — Go

- Designed a modular Go trading framework supporting identical strategy logic across backtesting and live execution through interchangeable clock, scheduler, execution, and portfolio components
- Built reusable components for market data, order execution, portfolio management, risk controls, scheduling, and historical backtesting

Market Data Service — Python, FastAPI

- Built a provider-agnostic market data service exposing historical OHLCV and ETF constituent data across 15+ years through REST APIs
- Implemented Parquet-based caching and scheduled constituent snapshots to reduce repeated upstream data requests and support reproducible historical backtests

Inclusive Search – Google HelloAI 2023 (internal) hackathon | Finalist

- Developed LLM-based prototype to diversify search results using cultural context embeddings
- The solution was ranked among the top 10 teams in the hackathon

SKILLS & CERTIFICATIONS

Programming Languages: Go, Python, C++, Java, TypeScript, JavaScript, SQL

Technologies: Redis, ClickHouse, PostgreSQL, MySQL, MongoDB, React, Next.js, Docker